

Unit III

Error Detection and Correction

Course Outcomes(CO)

CO3 - Analyze the Transmission Errors with respect to IEEE standards.

Theory Learning Outcomes(TLO's)

TLO 3.1 Explain working of the given error detection and correction method.

TLO 3.2 Explain features of the given IEEE communication standard.

TLO 3.3 Explain characteristics of the given layer in IEEE 802.11 architecture.

TLO 3.4 Explain with sketches the process of creating a Bluetooth environment using the given architecture.

TLO 3.5 Compare the specified generations of mobile telephone systems on the given parameter.

3.1 Types of errors:

1. Content errors

- The Content errors are nothing but errors in the contents of a message
- e.g. a "0" may be received as "1" or vice versa. Such errors are introduced due to noise added into the data signal during transmission.

2. Flow integrity errors.

- Flow integrity errors means missing blocks of data. It is possible that a data block may be Lost in the network as it has been delivered to a wrong destination.
- Depending on the number of bits in error we can classify the errors into two types as,

Single bit error

- In this type of error only one bit in the given data unit such as byte is in error.
- That means only one bit has been changed from 1 to 0 or 0 to 1, as shown in Fig

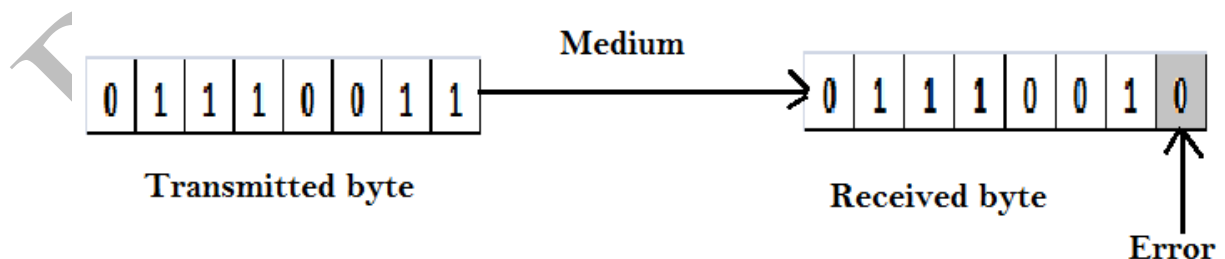


Fig 3.1: Single bit error

Burst Error:-

- If two or more bits from a data unit such as a byte change from 1 to 0 or from 0 to 1 then Burst error are said to have occurred .
- The Length of the burst is measured from the first corrupted bit to the last corrupted bit. Some of the bits in between may not have been corrupted.

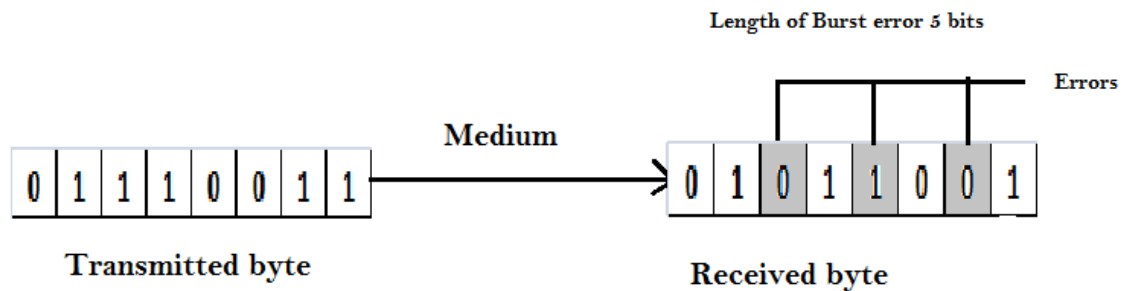


Fig 3.2: Burst Error

Forward Error Correction (FEC) Vs Re-Transmission

Forward Error Correction is a technique used to minimize errors in data transmission over communication channels by using some technique at receiver side which could correct the error or reproduce the packet immediately and give the receiver the ability to correct errors without needing the sender of the data to re-transmit the data. Because of this approach it saves bandwidth that is wasted for re-transmission. The receiver does not wait for the sender to re-transmit the data. In this method, the sender sends a redundant error-correcting code along with the data frame. The receiver performs necessary checks based upon the additional redundant bits. If receiver finds that the data is free from errors, it executes error-correcting code that generates the actual frame. It then removes the redundant bits before passing the message to the next upper layer.

Error correcting codes for forward error corrections can be broadly categorized into two types, namely, block codes and convolution codes.

- **Block codes** – The message is divided into fixed-sized blocks of bits to which redundant bits are added for error correction.
- **Convolutional codes** – The message comprises of data streams of arbitrary length and parity symbols are generated by the sliding application of a Boolean function to the data stream.

There are four popularly used error correction codes.

- **Hamming Codes** – It is a block code that is capable of detecting up to two simultaneous bit errors and correcting single-bit errors.

- **Binary Convolution Code** – Here, an encoder processes an input sequence of bits of arbitrary length and generates a sequence of output bits.
- **Reed - Solomon Code** – They are block codes that are capable of correcting burst errors in the received data block.
- **Low-Density Parity Check Code** – It is a block code specified by a parity-check matrix containing a low density of 1s. They are suitable for large block sizes in very noisy channels.

Advantages:-

- 1) FEC does not require handshaking between the source and the destination.
- 2) It can be used for broadcasting of data to many destinations simultaneously from a single source.
- 3) FEC saves bandwidth required for retransmission. So, it can be used in real time systems.

Disadvantages:-

- 1) If there are too many errors, the frames need to be retransmitted.

Re-Transmission:-

In Re-transmission sender re-transmits data if receiver encounter any error in the received data. In this mechanism sender sends the data and waits for acknowledgement. These techniques also use timer for retransmission. If Receiver finds any error in the data it sends negative acknowledgement for that data unit. The sender re-transmits the data for which negative acknowledgement is received or for which it has not received the acknowledgement.

Advantages:-

- 1) It Provides reliability of data.

Disadvantages:-

- 1) Require handshaking between the source and the destination.
- 2) It cannot be used for broadcasting of data to many destinations simultaneously from a single source.
- 3) More bandwidth is required for retransmission.

3.2 Framing:

The data link layer, needs to pack bits into frames, so that each frame is distinguishable from another. The DataLink layer prepares a packet for transport across the local media by encapsulating it with a header and a trailer to create a frame.

The Data Link layer frame includes:

- Data - The packet from the Network layer
- Header - Contains control information, such as addressing, and is located at the beginning of the PDU
- Trailer - Contains control information added to the end of the PDU

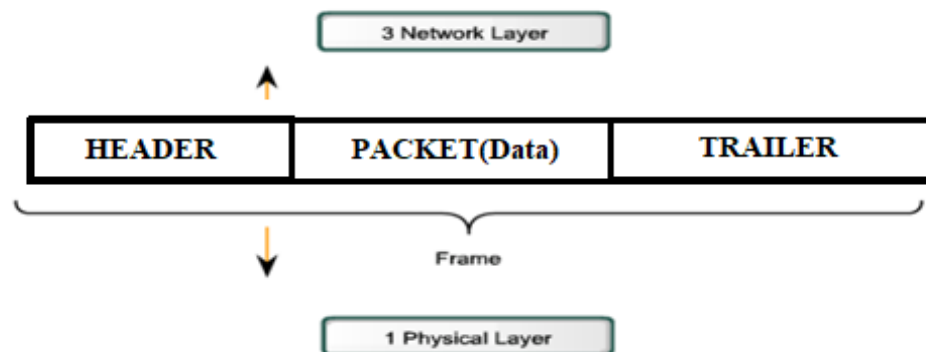


Fig 3.3: Framing

3.2.1 Fixed-Size Framing:

In fixed-size framing, there is no need for defining the boundaries of the frames the size itself can be used as a delimiter. The size of the frame is fixed and so the frame length acts as delimiter of the frame. Consequently, it does not require additional boundary bits to identify the start and end of the frame.

3.2.2 Variable-Size Framing:

In variable-size framing, we need a way to define the end of the frame and the beginning of the next.

1) Character Count Method

The first framing method uses a field in the header to specify the number of characters in the frame. When the data link layer at the destination sees the character count, it knows how many characters follow and hence where the end of the frame is. This technique is shown in Fig 3.4 for four frames of sizes 5, 5, 8, and 8 characters, respectively.

Figure 3.4 A character stream. (a) Without errors. (b) With one error.

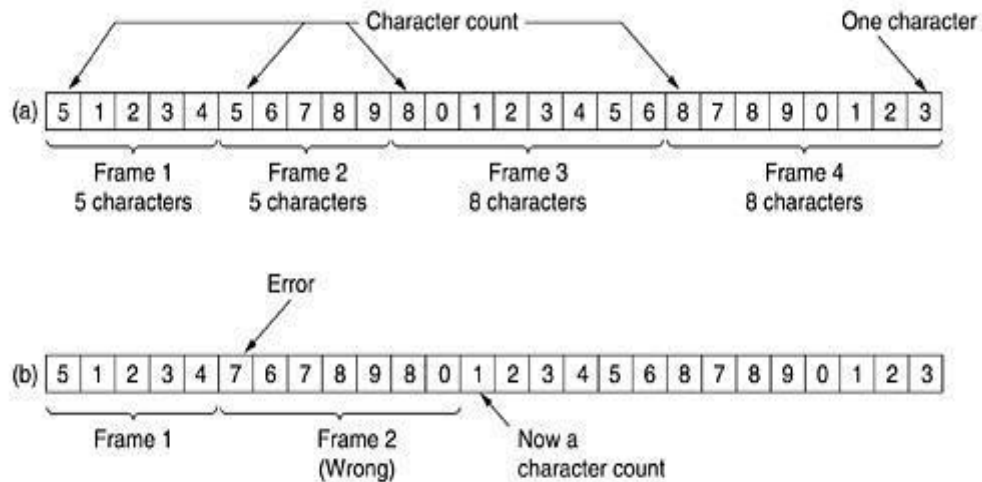


Fig 3.4 : character stream. (a) Without errors. (b) With one error.

Disadvantage:-

The trouble with this algorithm is that the count can be garbled by a transmission error. For example, if the character count of 5 in the second frame becomes a 7, the destination will get out of synchronization and will be unable to locate the start of the next frame. Even if the checksum is incorrect so the destination knows that the frame is bad, it still has no way of telling where the next frame starts. Sending a frame back to the source asking for a retransmission does not help either, since the destination does not know how many characters to skip over to get to the start of the retransmission. For this reason, the character count method is rarely used anymore.

2) Starting and Ending Character with character Stuffing

- Operates on bytes instead of bits. Use reserved characters to indicate the start and end of a frame.
- For instance, use the two-character sequence DLE STX (Data-Link Escape, Start of TeXt) to signal the beginning of a frame, and the sequence DLE ETX (End of TeXt) to flag the frame's end. Problem: What happens if the two-character sequence DLE ETX happens to appear in the frame itself?
- Solution: Use *character stuffing*; within the frame, replace every occurrence of DLE with the two-character sequence DLE DLE. The receiver reverses the processes, replacing every occurrence of DLE DLE with a single DLE.

Example:

If the frame contained "A B DLE D E DLE", the characters transmitted over the channel would be "DLE STX A B DLE DLE D E DLE DLE DLE ETX".

3) Starting and Ending Flags with Bit Stuffing

In this data frames contains an arbitrary number of bits and allows character codes with an arbitrary number of bits per character.

- It works like this. Each frame begins and ends with a special bit pattern, **01111110** (in fact, a flag byte).
- Whenever the sender's data link layer encounters **five** consecutive **1s** in the data, it automatically stuffs a **0 bit into the outgoing bit stream**.
- This **bit stuffing** is analogous to byte stuffing, in which an escape byte is stuffed into the outgoing character stream before a flag byte in the data.
- When the receiver sees five consecutive incoming 1 bits, followed by a 0 bit, it automatically destuffs (i.e., deletes) the 0 bit.
- Just as byte stuffing is completely transparent to the network layer in both computers, so is bit stuffing.
- **If the user data contain the flag pattern, 01111110, this flag is transmitted as 011111010 but stored in the receiver's memory as 01111110.**

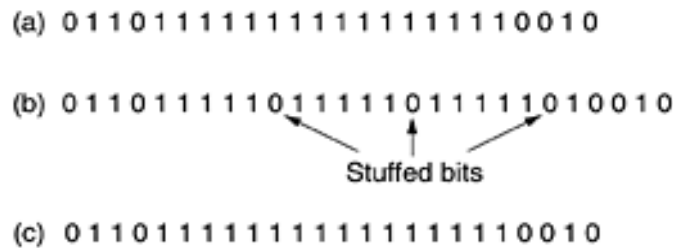


Fig 3.5: Bit Stuffing

3.3 Error Detection:-

- For error detection and correction it is necessary to add some check bit to a block of data bits.
- These check bits are also known as redundant bits because they do not carry any useful information.

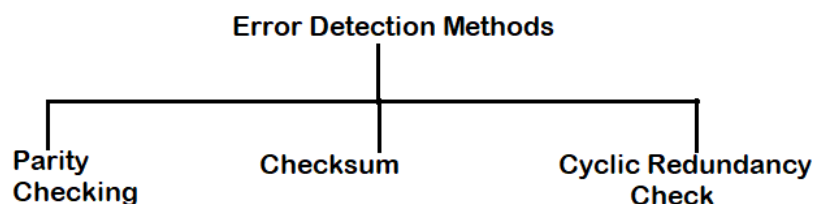


Fig 3.6: Error Detection Methods

3.3.1 Parity :

- In this technique an extra bit known as parity bit is added to each word being transmitted for detecting errors.
- As shown in Fig. generally the MSB of an 8-bit word is used as the parity bit and the remaining 7 bits are used as data or message bits.

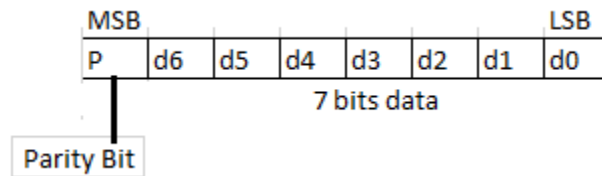
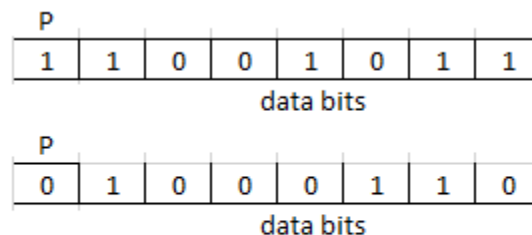


Fig 3.7: Parity Bit

- The parity of the 8-bit transmitted word can be either even parity or odd parity.
- Even parity means the number of 1's in the given word including the parity bit should be even (2, 4, 6..).
- Odd parity means the number of 1's in the given word including the parity bit should be odd (1, 3, 5 ...)

Use of parity bit to decide parity:

- The parity bit can be set to 0 or 1 depending on the type of parity required.
- For odd parity this bit is set to 1 or 0 at the transmitter such that the number of "1 bits" in the entire word is odd.



Inclusion of parity bit to obtain odd parity

Fig 3.8: Odd Parity

For even parity this bit is set to 1 or 0 such that the number of "1 bits" in the entire word is even.

P								
0	1	0	0	1	0	1	1	
data bits								

P								
1	1	0	0	0	1	1	0	
data bits								

inclusion of parity bit to obtain even parity

Fig 3.9: Even Parity

How does error detection take place?

- The parity checking at the receiver can detect the presence of an error if the parity of the received signal is different from the expected parity.
- That means if it is known that the parity of the transmitted signal is always going to be even and if the received signal has an odd parity then the receiver can conclude that the received signal is not correct.
- When a single error or an odd number of errors occur during transmission the parity of the codeword changes.
- Parity of the received codeword is checked at the receiver and change in parity indicates that error is present in the received word.
- if presence of error is detected then the receiver will ignore the received byte and request for Re-transmission of the same byte to the transmitter.

	Message bits								Parity	Receivers Decision
Transmitted Code	P									
	0	1	0	0	1	0	1	1	0	Even
Received Code with one error	P									
	0	0	0	0	1	0	1	1	0	Odd
Received Code with three errors	P									
	0	0	1	0	0	0	1	1	0	Odd

Fig 3.10: Error Detection using Odd Parity

When does parity checking fail to detect errors?

If the number of errors introduced in the transmitted code is two or any even number, then the parity of the received codeword will not change. It will still remain even as shown in Fig. and the receiver will fail to detect the presence of errors.

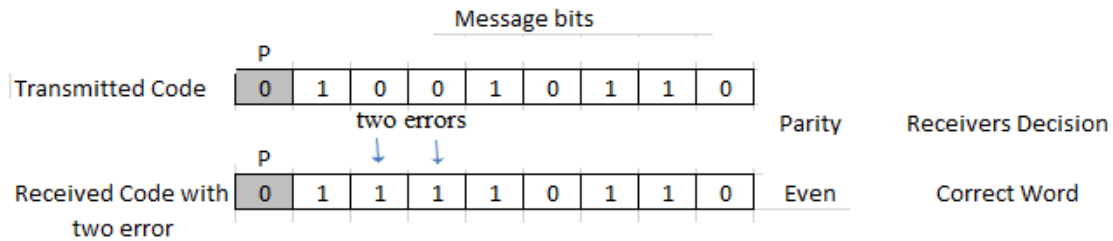


Fig 3.11: Multi Bit Error

Limitations of parity checking:

- The simple parity checking method has its limitations. It is not suitable for detection of multiple errors (two, four, six etc).
- The other limitation of parity checking method is that it cannot reveal the location of error bit. It cannot correct the error either.

3.3.2 Two dimensional Parity Check

When a large number of binary words are being transmitted or received in succession, the resulting collection of bits is considered as a block of data, with rows and columns

- The parity bits are produced for each row and column of such block of data.
 - The two sets of parity bits are generated:-
 - Longitudinal Redundancy Check (LRC) bits
 - Vertical Redundancy Check (VRC) bits.
- the LRC bits indicate the parity of rows & VRC bits indicate parity of columns as in fig. 3.12

Characters	E	L	E	P	H	A	N	T		
Message Bits	1	1	0	1	1	0	1	1	0	
	1	0	0	1	1	0	1	1	1	
	1	1	1	1	1	0	1	1	1	
	1	0	0	1	0	0	1	1	0	
	1	1	0	1	1	0	0	0	0	
	1	1	0	0	0	0	1	1	0	
	1	1	0	1	1	0	1	0	1	
VRC bits	1	1	1	0	1	0	0	1	1	
										LRC Bits

Fig 3.12: Two Dimensional Parity

The Longitudinal Redundancy Check (LRC) :

The LRC bits are parity bits associated with the rows of the data block. Each LRC bit will make the parity of the corresponding row, an even parity. For example, consider row 1 fig 3.13

Row 1	1	1	0	1	1	0	1	1	0	LRC Bit to make Parity Even
-------	---	---	---	---	---	---	---	---	---	-----------------------------

Fig 3.13: Longitudinal Redundancy Check

The Vertical Redundancy Check (VRC) :

As shown in Fig the VRC bits are parity bits associated with the ASCII code of each character. Each VRC bit will make the parity of its corresponding column "an even parity". For example consider column 1 corresponding to character "C". The ASCII code for the character C is

Characters	E
Message Bits	1
	1
	1
	1
	1
	1
	1
VRC bits	1

VRC Bit 1 to make parity of first cloumn

Fig 3.14: Vertical Redundancy Check

Therefore the 8th bit which is a VRC bit is made "1" to make the parity even.

How to locate the bit in error?

- Even a single error in any bit will result in a non correct "LRC" in one of the rows and an incorrect VRC in one of the columns. The bit which is common to the row and column is the bit in error. however there is still a limitation on the Block parity code, which is that, multiple errors in rows and columns can be only detected but the cannot be corrected. This is because it is not possible to locate the bits which are in error.

The following bit stream is encoded using VRC, LRC and even parity. Locate and correct the error if it is present.

```

11000011 11110011 10110010 00001010
00101010 00101011 10100011 01001011
11100001

```

Fig 3.15: Received Data Block

Fig 3.15 shows the received data block along with the LRC and VRC bits. Note the parity bits corresponding to row 1 and column 5 indicate wrong parity. Therefore the fifth bit in the first row (encircled bit) is incorrect. Using VRC and LRC, it is possible to locate and correct the bits in error.

					Bit in Error					LRC bits(Even Parity)
					↓					
	b1	1	1	1	0	0	0	1	0	1
	b2	1	1	0	0	0	0	0	1	1
	b3	0	1	1	0	1	1	1	0	1
data	b4	0	1	1	0	0	0	0	0	0
blocks	b5	0	0	0	1	1	1	0	1	0
	b6	0	0	0	0	0	0	0	0	0
	b7	1	1	1	1	1	1	1	1	0
VRC bits		1	1	0	0	0	1	1	1	1
Even Parity										
					↑					Wrong Parity

Fig 3.16: Error Identification using LRC & VRC

3.3.3 Checksum for error detection :

- simple parity cannot detect two or even number of errors within the same word.
- one way to overcome this problem is to use a sort of two dimensional parity.
- As each word is transmitted, it is added to the previously sent word and the sum is retained at the transmitter as shown in Fig

Word A :	1 0 1 1 0 1 1 1
	+
Word B :	0 0 1 0 0 0 1 0
Sum :	1 1 0 1 1 0 0 1

Fig 3.17: Check sum Method

- Each successive word is added in this manner to the previous sum. At the end of the transmission the sum (called a checksum) upto that time is sent. ,
- The errors normally occur in burst. The parity check method is not useful in detecting the errors under such conditions. The checksum error detection method can be used successfully in detecting such errors. ,
- In this method a "checksum" is transmitted along with every block of data bytes. In this method an eight bit accumulator is used to add 8 bit bytes of a block of data to find the "checksum byte". The carries of the MSB are ignored while finding out the checksum byte.

How to detect error using the checksum byte?

- After transmitting a block of data bytes (e.g 8-data bytes) the "checksum" byte is also transmitted. The checksum byte is regenerated at the receiver separately by adding the received bytes.
- The regenerated checksum byte is then compared with the transmitted one. If both are identical then there is no error. If they are different then the errors are present in the block of received data bytes.
- Sometime the 2's complement of the checksum is transmitted instead of the checksum itself. The receiver will accumulate all the bytes including the 2's complement of the checksum.
- If there is 0 error, the contents of the accumulator should be zero after accumulation of the 2's complement of the checksum.

Advantage of the checksum method :

The advantage of this method over the simple parity checking method is that the data bits are "mixed up" due to the 8 bit addition. Therefore checksum represents the overall data block. In checksum there ore, there is 255 to 1 chance of detecting random errors.

3.3.4 Cyclic Redundancy Check (CRC) :

- This is a type of polynomial code in which a bit string is represented in the form of polynomials with coefficients of 0 and 1 only.
- Polynomial arithmetic uses a modulo-2 arithmetic i.e. addition and subtraction are identical to XOR.

- For CRC code the sender and receiver should agree upon a generator polynomial $G(x)$. A code word can be generated for a given data word (message) polynomial $M(x)$ with the help of long division.
- CRC is based on binary division: A sequence of redundant bits called CRC or CRC remainder is appended at the end of a data unit such as byte. -
- The resulting data unit after adding CRC remainder becomes exactly divisible by another predetermined binary number.
- At the receiver, this data unit is divided by the same binary number.
- There is no error if the remainder of division zero.
- But a non-zero remainder indicates presence of errors in the received data unit.
- Such an erroneous data unit is then rejected

Procedure to obtain CRC:

1. Divide the data unit by a predetermined divisor.
2. Obtain the remainder. It is the CRC.

Requirement of CRC:-

The CRC will be valid only if it satisfies the following requirements:

1. It should have exactly one bit less than divisor.
2. Appending the CRC to the end of the data unit should result in the bit sequence which is exactly divisible by the divisor.

CRC generator :

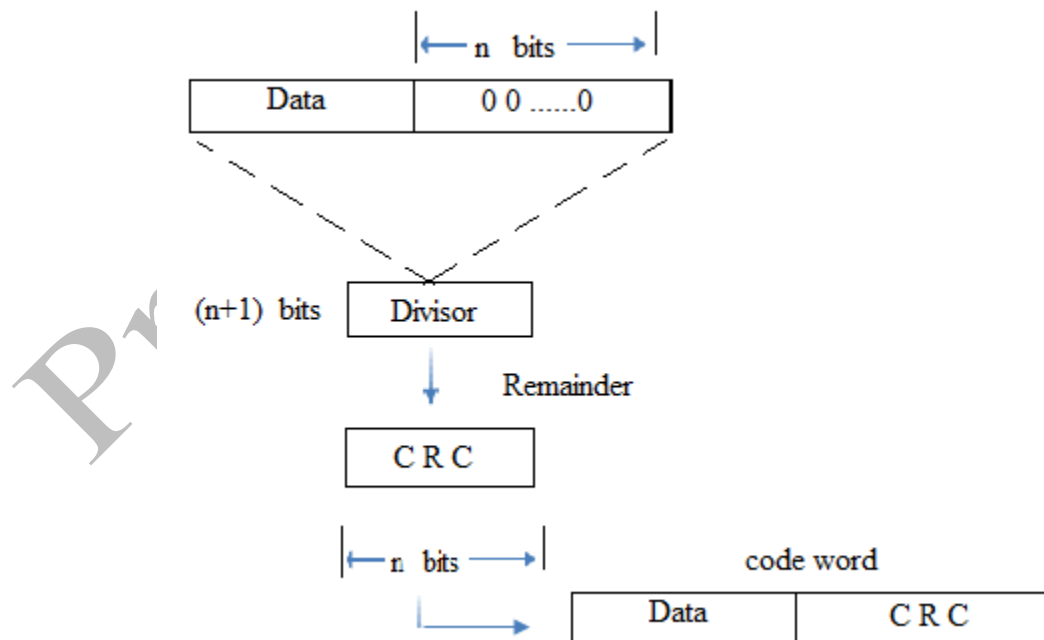


Fig 3.18: CRC generator

Step 1: append a string of n 0's to the data unit where n is 1 less than the number of bits in the Pre-decided divisor (n + 1 bit long).

Step 2: Divide the newly generated data unit in step 1 by the divisor. This is a binary division.

Step 3 : the remainder obtained after the division in step 2 is the n bit CRC

Step 4: This CRC will be appended to the data unit , to get the codeword to be transmitted as shown in Fig.3.18

Mod-2 Addition:

- The rules of modulo - 2 addition are as follows.

$$0 \oplus 0 = 0$$

$$0 \oplus 1 = 1$$

$$1 \oplus 0 = 1$$

$$1 \oplus 1 = 0$$

CRC checker :

- The codeword received at the receiver consists of data and CRC.
- The receiver treats It as one unit and divides It by the same (n + 1) bit divisor which was used at the transmitter.
- The remainder of this division is then checked.

If the remainder is zero, then the received codeword is error free and hence should be accepted.

- But a non-zero remainder indicates presence of errors hence the corresponding codeword should be rejected.

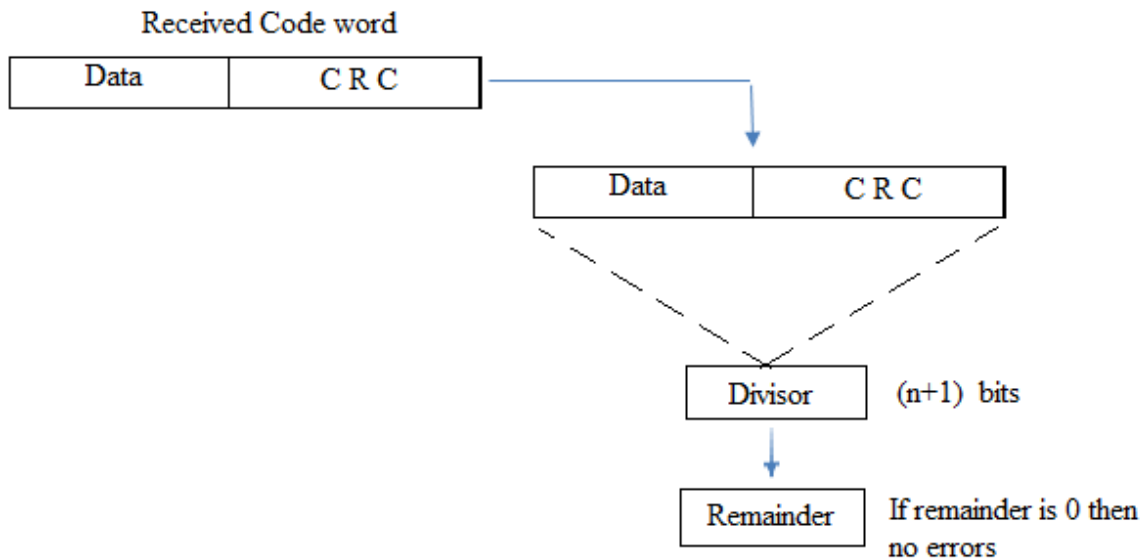


Fig 3.19: CRC checker

3.4 Error Correction

Recovery from errors:-

There are various techniques for error correction. One of them is to send some kind of an acknowledgement from the receiver to sender. The receiver will send a positive acknowledgement (ACK) to the sender after receiving a correct packet. But if the receiver receives an erroneous packet or if the packet is lost, then it will send a Negative Acknowledgement (NAK) to the sender. Sender will always wait until it receives an ACK or NAK. If it receives the ACK, then it decides to send the next block of data. But if a NAK is received, then the sender will retransmit the same block of data.

Automatic repeat request (ARQ)

- Also known as **automatic repeat query**, is an error-control method for data transmission that uses acknowledgments (messages sent by the receiver indicating that it has correctly received a packet) and timeouts (specified periods of time allowed to elapse before an acknowledgment is to be received) to achieve reliable data transmission over an unreliable service.
- If the sender does not receive an acknowledgment before the timeout, it usually retransmits the packet until the sender receives an acknowledgment or exceeds a predefined number of retransmissions.

Stop & wait :-

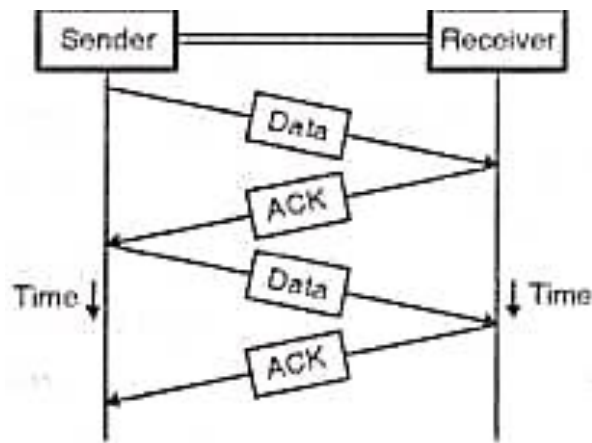


Fig 3.20: Stop & wait

In this method, the sender sends one data frame and then waits for an acknowledgement (ACK) from the receiver. Sender will send the next frame only after receiving the acknowledgement. The advantage of stop-and-wait approach is that every frame is acknowledged before the transmission of next frame. However this approach is slow. transmission rates with stop-and-wait is low.

Timer and time out(lost Frames):

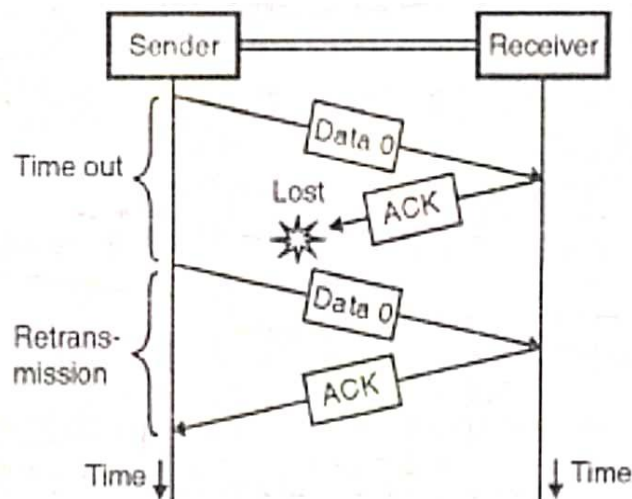


Fig 3.21 : Stop & Wait using time out

- The sender sets a timer for every transmitted frame. If the sender does not receive the acknowledgement within this set time, then it retransmits the same data frame again as shown in Fig 3.21

- This situation arises when the data Frame or acknowledgement frame is lost.
- If a data frame is lost, it won't reach the receiver. So sender does not receive any acknowledgement.
- Hence it will retransmit the same frame after the timer goes off .

Fig 3.22 shows the loss of data frame

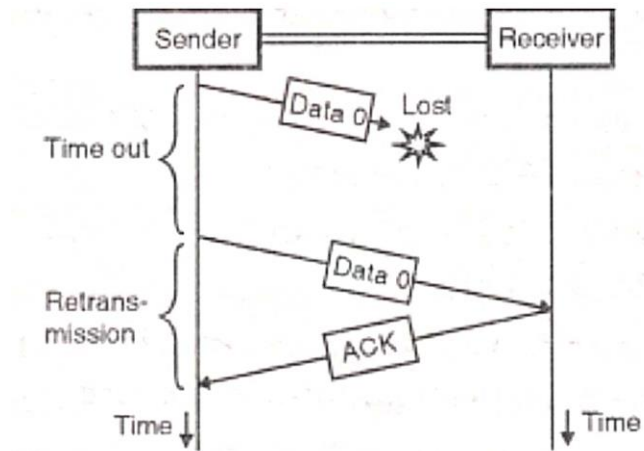


Fig 3.22: Stop and wait with data lost

Stop and wait with damaged frames:

- If there is an error in the received frame the receiver sends a NAK requesting retransmission of the data frame.

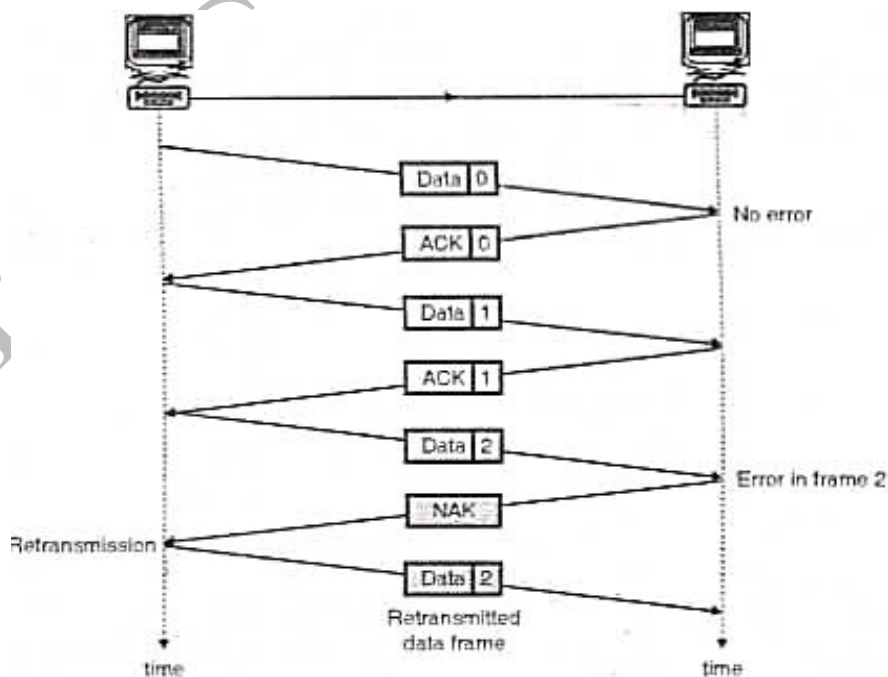


Fig 3.23: Stop and wait with damaged frame

Disadvantage of stop and wait protocol:

1. At any moment, only one frame is in transition.
2. The sender will have to wait at least one round trip time before sending next frame.

Go back n:-

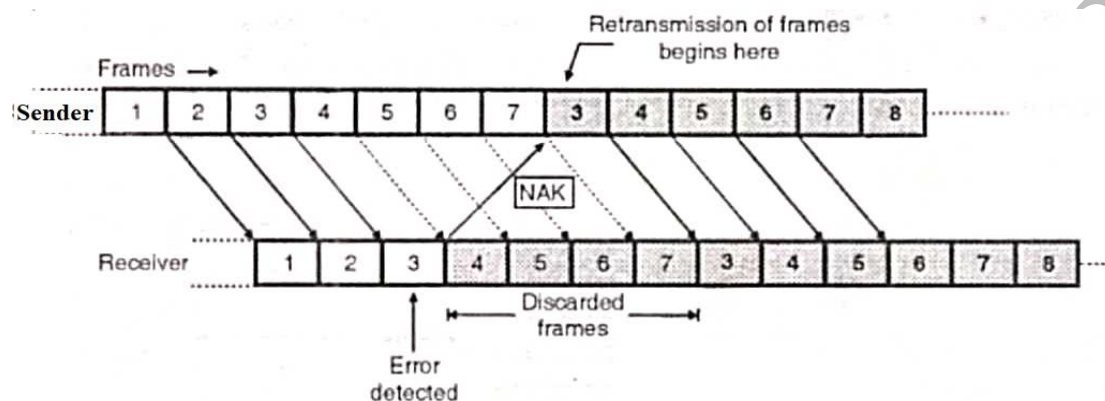


Fig 3.24: Go Back n

- In this Technique The sender sends more than one data frames at a time (upto 8 frames). If all these frames are received without any error, then the receiver sends an ACK signal for this group of frames.
- But if there is error in the frame then the receiver send a NAK signal back to the sender. Then the sender will retransmit all the frames again.
- It is the method used to overcome the inefficiency of the stop and wait by allowing the transmitter to continue sending enough frames so that the channel is kept busy while the transmitter waits for acknowledgements.
- In this method if one frame is damaged or lost, all frames are sent again from the last frame acknowledged

Principle of GO-back-n :

- The sender does not wait for ACK signal for the transmission of next frame.
- It transmits the frames continuously as long as it does not receive the "NAK" signal. NAK is the negative acknowledgement signal sent by the receiver to the transmitter.
- When the receiver detects an error in the third frame as shown in Fig 3.25 the receiver send a NAK signal back to sender.
- But the signal takes some time to reach the transmitter. By that time the transmitter has transmitted frames up-to frame 7.

- On reception of the NAK signal, the transmitter will retransmit all the frames from 3 onwards. The receiver discards all the frames it has received after 3 i.e. 3 to 7. It will then receive all the frames that are retransmitted by the transmitter.
- Now the sender should transmit all the frames from 3 onwards.

Operation when a frame is damaged:

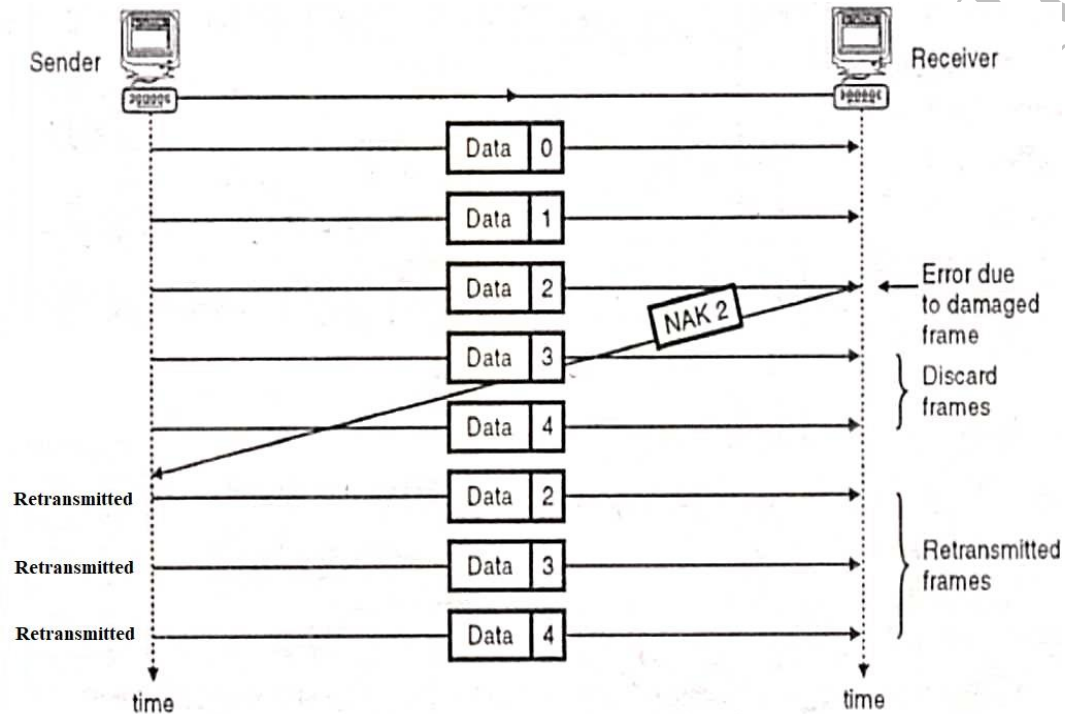


Fig 3.25: frame is damaged

- The second data frame is damaged, so the error is detected and receiver send NAK-2 signal To the transmitter.
- On receiving this signal, the transmitter starts retransmission from frame 2.
- All the frames received after frame 2 are discarded by the receiver.

Operation when a frame is lost:

- The case of lost frame is also treated in the same manner as that of the damaged frame.

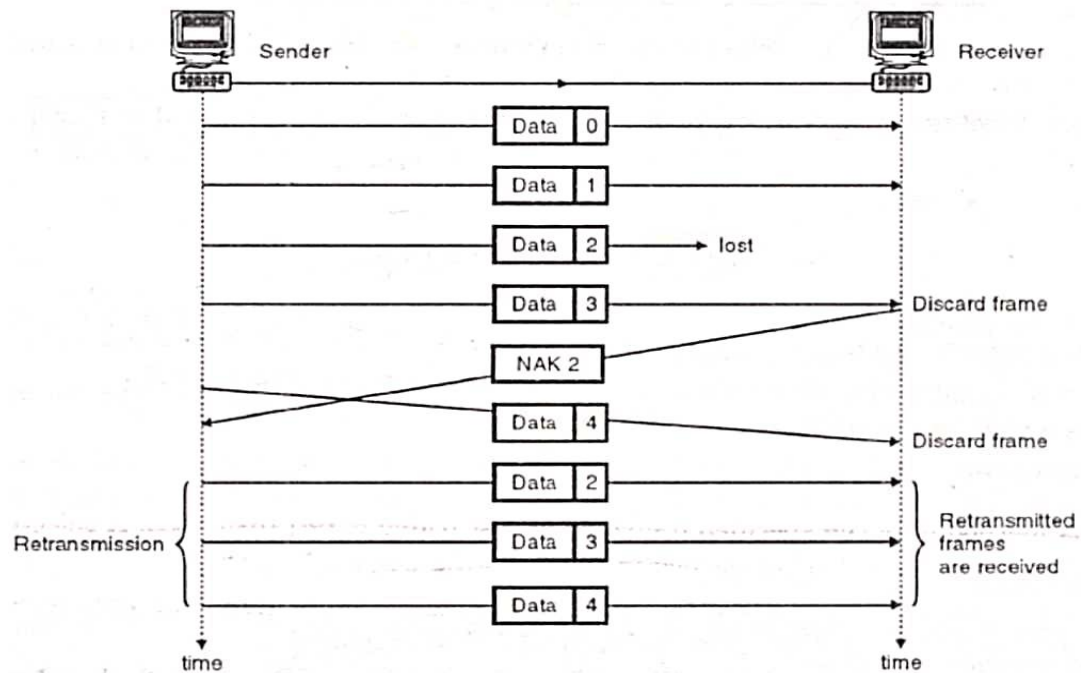


Fig 3.26: Frame is lost

- The receiver, if it does not receive a particular data frame it sends a NAK to the transmitter and the transmitter retransmits all the frames, from the frame that was negative acknowledged.

Operation when the acknowledgement is lost:

- In case of go-back-n method the transmitter does not expect an acknowledgement after every data frame.

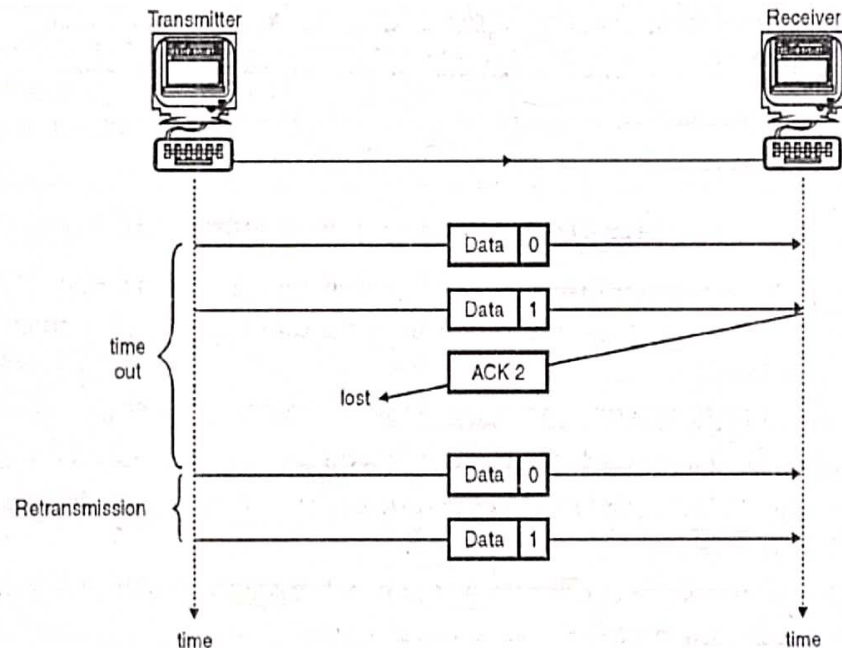


Fig 3.27: Acknowledgement is lost

- It cannot use the absence of sequential ACK numbers to identify lost ACK or NAK frames, instead it uses a timer.
- The transmitter can send as many frames as the window allows before waiting for an acknowledgement.
- Once the limit has been reached or the transmitter has no more frames to transmit it must wait till the timer goes off and retransmit all the data frames again.
- The disadvantage of Go-back-n ARQ protocol is that in noisy channels it has poor efficiency because of the need to retransmit the frame in error and all the subsequent frames.

Disadvantages of Go back n :

1. It transmits all the frames if one frame is damaged or lost.
2. It transmits frames continuously as long as it does not receive the NAK signal.
3. The NAK signal takes some time to reach the sender. Till that time the sender has already sent some frames. All those will be retransmitted after receiving the NAK.
4. The error can get introduced if the NAK is lost.

Selective Repeat Request:

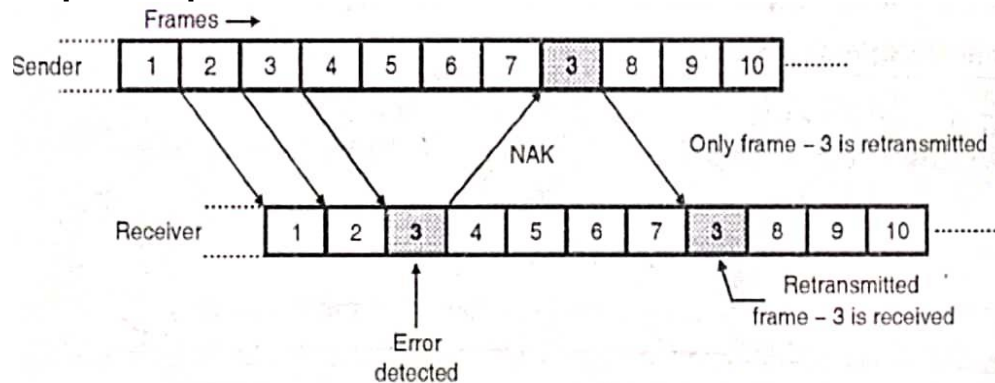


Fig 3.28: Selective Repeat Request

- In this method only the specified damaged or lost frame is retransmitted.
- The receiver can do sorting of data frames and is also able to store frames received after a NAK has been sent until the damaged frame has been replaced.
- The transmitter must contain a searching mechanism that allows it to find and select only the requested frame for retransmission.
- In this system as well, the transmitter does not wait for the ACK signal for the transmission of the next frame. It transmits the frame continuously till it receives the "NAK" signal from the receiver.
- The receiver sends the "NAK" signal to the transmitter as soon as it detects an error in the received frame. For example the receiver detects an error in the third frame.
- By the time this "NAK" signal reaches the transmitter, it had transmitted the frames up-to 7 as shown in Fig 3.28
- On reception of "NAK" signal, the transmitter will retransmit only the frame-3 and then Continues with the sequence 8,9 ...
- The frames 4, 5, 6 and 7 received by the receiver are not discarded by the receiver. The receiver receives the retransmitted frames in between the regular frames. Therefore the receiver will have to maintain the frames sequentially.
- The disadvantage of this method is that because of the complexity of sorting and storage required by the receiver and the extra logic needed by the transmitter to select frames for retransmission, the system becomes more expensive.
- The advantage of this system is that it gives the best throughput efficiency. This is due to the use of pipelining in selective repeat ARQ.

3.5 IEEE 802.11 WLAN Standards

- In 1997, IEEE approved 802.11, the First sanctioned WLAN standard.
- This First standard allowed three possible implementations for the physical layer: infrared (IR) pulse position modulation, or radio frequency (RF) signalling in the 2.4 GHz band using either Frequency Hopping Spread Spectrum (FHSS) or Direct Sequence Spread Spectrum (DSSS) .

- IEEE established two networking groups (A and B) to explore alternate implementations of 802.11. A third group, working group G, was set up after these two.

IEEE 802.11 Architecture

- The IEEE 802.11 standard defines two kinds of services:
the Basic Service Set (BSS) and the
Extended Service Set (ESS)

BSS-

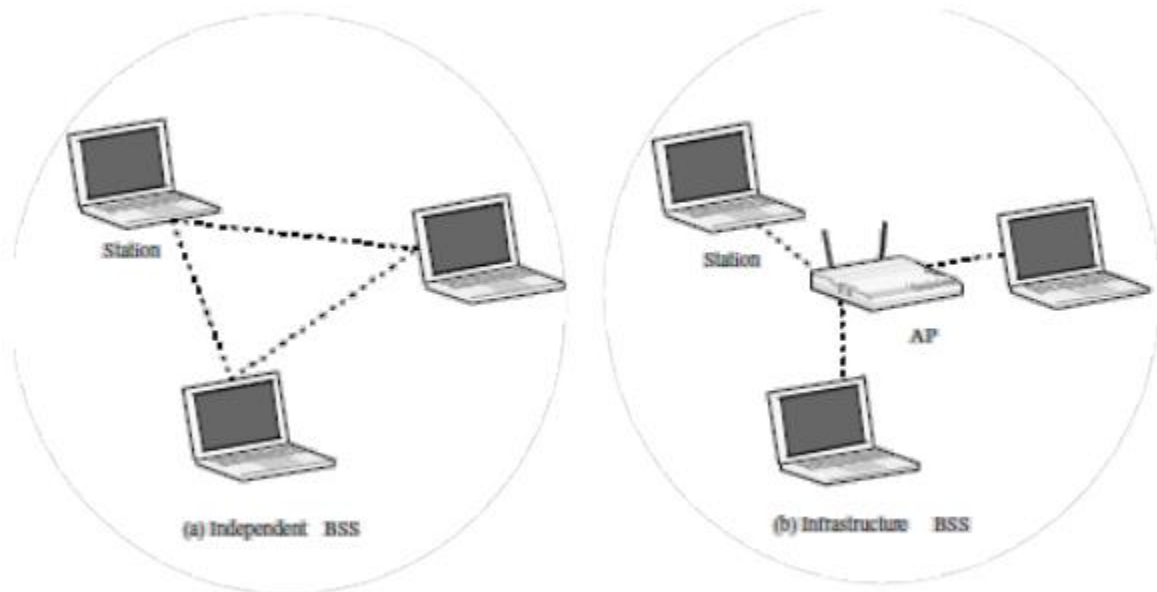


Fig 3.29 : BSS

- The BSS is the basic building block of a wireless LAN. A BSS consists of stationary or mobile wireless stations and possibly a central base station (e.g., an AP).
- When a station is in the BSS, it can communicate with the other members of the BSS.
- The BSS without an AP is a stand-alone network and cannot send data to other BSSs. Such BSSs are called Independent BSSs (IBSS).
- Typically, IBSSs involve a small number of stations set up for a specific purpose and for a short period of time (e.g., creating a short-lived network to support a single meeting in a conference room).
- IBSSs are also referred to as ad hoc networks.
- Infrastructure BSSs are distinguished from ad hoc networks by the use of an AP.
- APs are used for all communications in an infrastructure BSS, including communication between mobile nodes in the same service set.
- An infrastructure BSS is bounded by the coverage distance from the AP.

- The coverage area of a single AP is called a cell. All mobile stations are required to be within reach of the AP. 802.11 allows wireless networks of arbitrarily large size to be created by linking BSSs into an ESS.

ESS-

- An ESS is created by chaining BSSs together with a backbone network. All the APs in an ESS are given the same Service Set Identifier (SSID), which serves as a network name for its users.
- APs in an ESS operate in a manner such that the outside world can use the station's MAC address to talk to a station without worrying about its location in the ESS.
- Figure shows three BSSs corresponding to three APs. There is an equal level of overlap between BSS 1 and BSS 2, and between BSS 2 and BSS 3. Such overlap is necessary to provide stations with seamless connectivity if they move from one BSS to another.
- In the figure, the router uses the station's MAC address as the destination to deliver frames to a station; only the AP with which that station is associated delivers the frame.
- IEEE 802.11 allows mobility between BSSs at the link layer. However, it is not aware of anything that happens above the link layer. When stations move between BSSs, they will find and attempt to associate with an AP with the strongest signal and the least network.

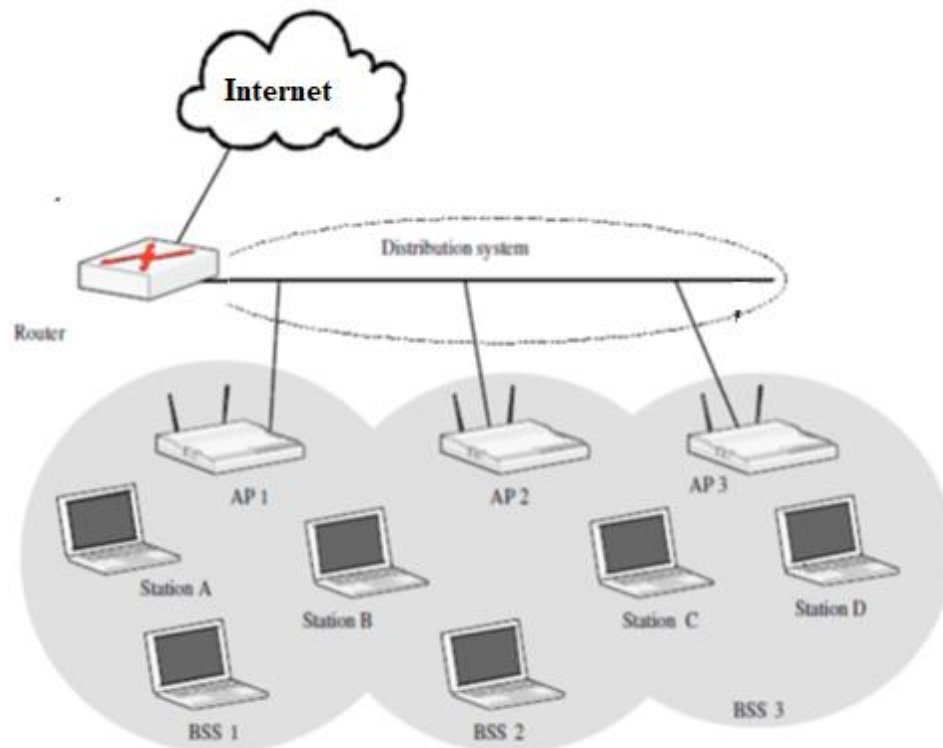


Fig 3.30: ESS

IEEE 802.11 Framing

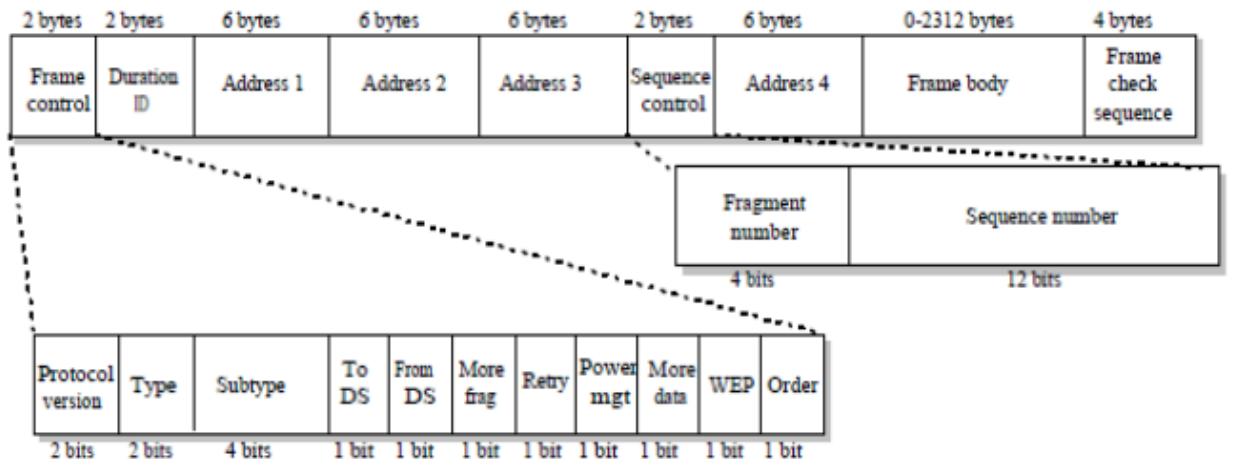


Fig 3.31 : IEEE 802.11 Frame

- 802.11 framing is complex compared to Ethernet framing. This is because the wireless medium requires several management features and frame types not found in wired networks.
- **Frame control** The frame control is 2 bytes long and defines the type of the frame and control information. The subfields in the frame control field are as follows:
- **Protocol version:** Indicates which version of 802.11 MAC is contained in the rest of the frame. Only one version of the 802.11 MAC has been developed: it has been assigned a protocol number of 0.
- **Type:** Defines the type of information carried in the frame body: management (00), control (01), or data (10).
- **ToDS:** This bit is set to 1 if the frame was sent to the DS.
- **FromDS:** This bit is set to 1 if the frame was sent from the DS.
- **More fragments:** When a higher-level packet has been fragmented by the MAC, the initial fragment and any following non-final fragments set this bit to 1.
- **Retry:** The retry bit is set to 1 if the current packet is a retransmission of a previous attempt. This aids the receiving station in eliminating duplicate packets.
- **Power management:** To conserve battery life, many small devices have the ability to power down parts of the network interface. A 1 indicates that the station will be in power save mode, and 0 indicates that the station will be active.
- **More data:** To accommodate stations in a power saving mode, APs may buffer frames received from the DS. An AP sets this bit to indicate that at least one frame addressed to a sleeping" station is available.
- **WEP:** Wireless transmissions are inherently easier to intercept than transmissions on a wired network. The Wired Equivalent Privacy (WEP) bit is set to 1 if the payload of the packet has been encrypted using the WEP algorithm.

- **Order:** Frames and fragments can be transmitted in any order by both the receiving and sending stations. The bit is set to 1 when the packets must be strictly ordered, for example, for VoIP.
- **Duration ID** The duration field is used to set the Network Allocation Vector (NAV). NAV is used for carrier sensing. The value represents the number of microseconds that the medium is expected to remain busy for the transmission currently in progress. In case of power saving stations, they create an association ID telling the AP their BSSs and where to send the buffered packets.
- **Address fields** There are four address fields, each 6 bytes long.

ToDS	FromDS	Address 1 (receiver)	Address 2 (transmitter)	Address 3	Address 4
0	0	Destination	Source	BSSID	N/A
0	1	Destination	Sending AP	Source	N/A
1	0	Receiving AP	Source	Destination	N/A
1	1	Receiving AP	Sending AP	Destination	Source

Sequence control The sequence control field is composed of a 4-bit fragment number and a 12-bit sequence number. This field is used for defragmentation and discarding duplicate frames. Higher-level frames are each given a sequence number as they are passed to the MAC layer for transmission. The sequence number subfield operates as a modulo-4096 counter of the frames transmitted. It begins at 0 and increments by 1 for each higher-level packet handled by the MAC. If higher-level packets are fragmented, all fragments will have the same sequence number. The first fragment is given a fragment number of 0 and successive fragments have their fragment number subfield incremented by 1. When frames are retransmitted, the sequence number is not changed.

Frame body The frame body moves the higher-layer payload from station to station. 802.11 can transmit frames with a maximum payload of 2304 bytes of data. On IP networks, path maximum transmission unit (MTU) typically prevents the transmission of frames with data larger than 1500 bytes. IEEE 802.11 does not generally pad frames to a minimum length.

Frame check sequence Similar to Ethernet, the 802.11 frame ends with a Frame Check Sequence (FCS), also known as Cyclic Redundancy Check (CRC). The FCS allows stations to check the integrity of received frames.

IEEE 802.11.a :-

- Group A explored the 5 GHz band, using Orthogonal Frequency Division Multiplexing (OFDM) to achieve transmission rates in the range of 54 Mbps.

IEEE 802.11.b :-

- Group B explored more sophisticated DSSS techniques in the original 2.4 GHz band.
- Their 802.11b WLAN standard, published in 1999, can deliver data rates up to 11 Mbps.

IEEE 802.11.g :-

- Group G began by exploring a variety of methods to further improve throughput in the 2.4 GHz spectrum used by the 802.11b standard.
- 802.11g standard adopting OFDM, the same signaling method used in the 802.11a WLAN standard.
- The 802.11g standard provides backward compatibility with the older 802.11b standard, which uses the same spectrum.
- Even though 802.11g operates in the same frequency band as 802.11b, it can achieve higher data rates because of its similarities to 802.11a.

Standard	Data rates (MBPS)
802.11a	6, 9, 12, 24, 36, 48, 54
802.11b	1, 2, 5.5, 11
802.11g	1, 2, 5.5, 6, 9, 11, 12, 22, 24, 36, 48, 54

3.6 Bluetooth

- Bluetooth is the name given to a new technology using short-range radio links, intended to replace the cable(s) connecting portable and/or fixed electronic devices.
- It is envisaged that it will allow for the replacement of the many propriety cables that connect one device to another with one universal radio link.
- Its key features are robustness, low complexity, low power and low cost.
- Designed to operate in noisy frequency environments, the Bluetooth radio uses a fast acknowledgement and frequency-hopping scheme to make the link robust.
- Bluetooth radio modules operate in the unlicensed ISM band at 2.4 GHz, and avoid interference from other signals by hopping to a new frequency after transmitting or receiving a packet.
- Compared with other systems in the same frequency band, the Bluetooth radio hops faster) and uses shorter wavelengths.
- Thus Bluetooth is a wireless LAN technology designed to connect devices of different functions such as telephones, computers, printers, cameras, etc.
- A Bluetooth LAN is an Ad hoc network. It is possible to connect the Bluetooth LAN to the Internet.
- This technology is implemented using the IEEE 802.15 standard.

Bluetooth defines two types of networks:

1. Piconets and 2. Scatternets.

- A Bluetooth network is called as a piconet or a small net. It can have at the most eight stations. One of them is called as a master and all others are called as slaves.
 - All the slave stations are synchronized with the master.
 - A piconet can have only one master station. Fig. 32 shows a piconet.
- The communication between a master and slaves can be one-to-one or one-to-many.

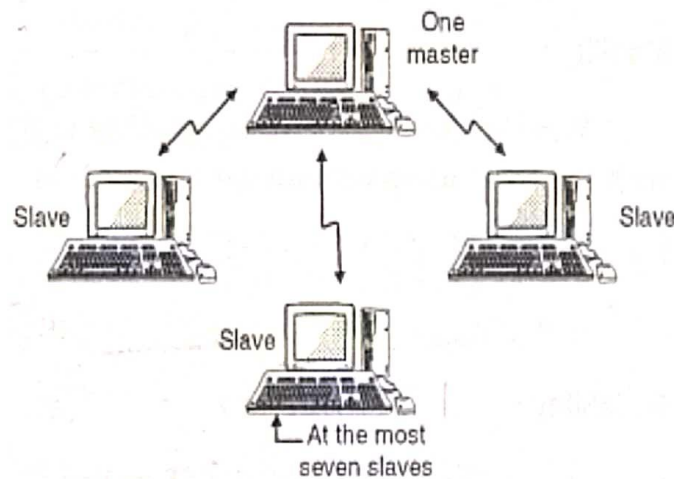


Fig 3.32: Piconet

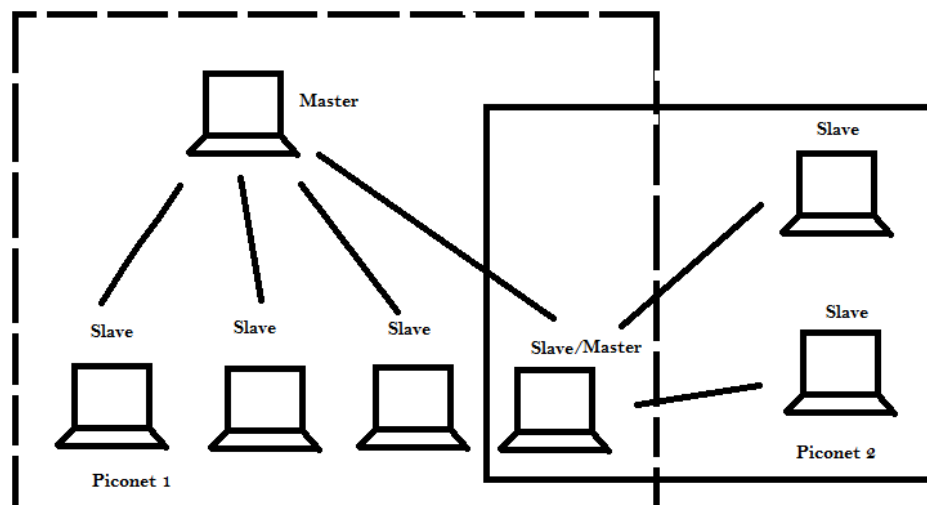


Fig 3.33: Scatternet

- Fig. 3.33 shows a scatternet consisting of two piconets. A slave in the first piconet can act as a master in the second piconet.
- It will receive the messages from the master in the first piconet by acting as a slave and then delivers the message to the, slaves in the second piconet as shown in Fig.

Bluetooth Devices:

- Every Bluetooth device consists of a built in short range **radio transmitter**. The current data rate is 1 Mbps and the bandwidth is 2.4 GHz.
- So an interface between the IEEE 802.11 wireless LAN and Bluetooth LAN is possible.
- The Bluetooth specification standard defines a short-range (10-meter) radio link.
- The devices carrying Bluetooth-enabled chips can easily transfer data at a rate of about 1Mbps (Megabits per second) within 10 meters (33 feet) of range through walls, clothing and luggage bags.
- The interaction between devices occurs by itself without direct human intervention whenever they are within each other's range. In this process, the software technology embedded in the Bluetooth transceiver chip triggers an automatic connection to deliver and accept the data flow.
- Since Bluetooth is of short range with limited speed and low-power technology. It is less attractive to corporate wireless local area networks that are generally powered with the 802.11 wireless LAN technologies.
- Each Bluetooth-enabled device contains a 1.5-inch square transceiver chip operating in the ISM (industrial, scientific, and medical) radio frequency band of 2.40 GHz to 2.48 GHz.
- This frequency is generally available world wide for free without any licensing restrictions. ISM band is divided into 79 channels with each carrying a bandwidth of 1 MHz.

Bluetooth Layers (Bluetooth Protocol Architecture):-

- The layers of Bluetooth as shown in Fig. 3.34 do not exactly match with the internet layers.

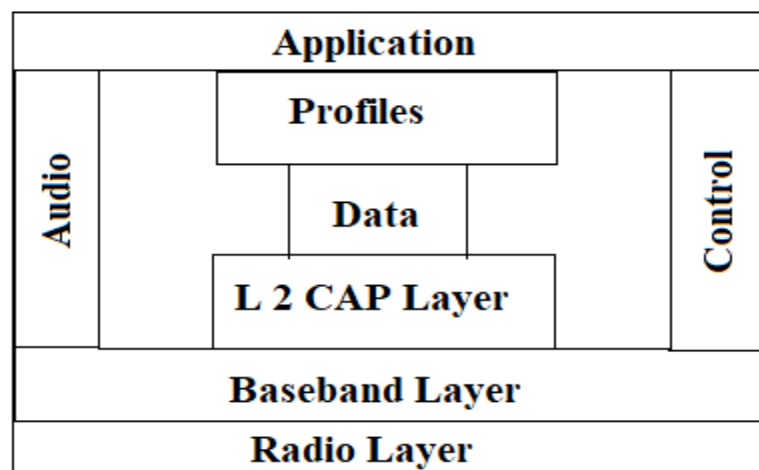


Fig 3.34 : Bluetooth Architecture

RADIO

The Bluetooth Radio (layer) is the lowest layer of the Bluetooth specification. It defines the requirements of the Bluetooth transceiver device operating in the 2.4 GHz ISM band. This band is divided into 79 channels of 1 MHz each.

Baseband

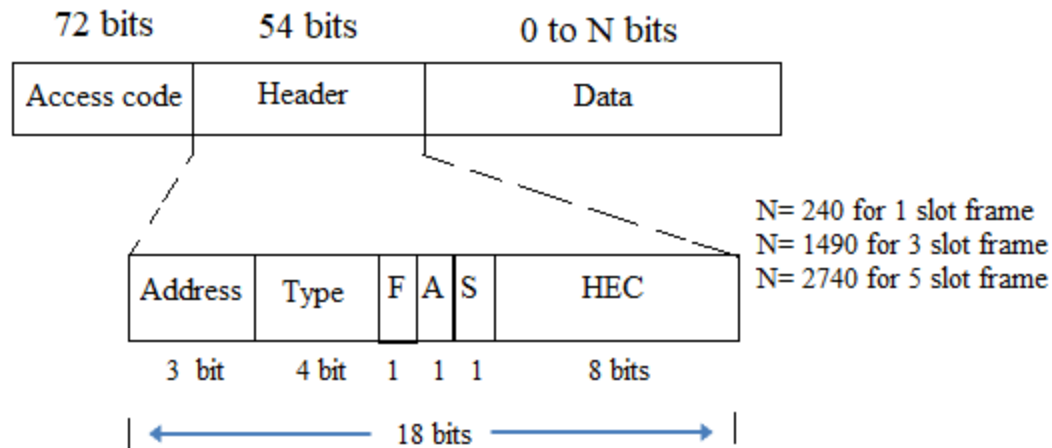
- The baseband is the physical layer of the Bluetooth. It manages physical channels and links apart from other services like error correction, data whitening, hop selection and Bluetooth security.
- The baseband layer lies on top of the Bluetooth radio layer in the bluetooth stack. The baseband protocol is implemented as a link controller, which works with the link manager for carrying out link level routines like link connection and power control.
- The baseband also manages asynchronous and synchronous links, handles packets and does paging and inquiry to access and inquire Bluetooth devices in the area.
- The baseband transceiver applies a Time-Division Duplex (TDD) scheme. (alternate transmits and receives).
- Therefore apart from different hopping frequency (frequency division), the time is also slotted.

Logic link Control and Adaptation protocol(L2CAP)

- The Logical Link Control and Adaptation Layer Protocol (L2CAP) is layered over the baseband protocol and resides in the data link layer.
- L2CAP provides connection-oriented and connectionless data services to upper layer protocols with protocol multiplexing capability, segmentation and reassembly operation, and group abstractions. L2CAP permits higher-level protocols and applications to transmit and receive L2CAP data packets up to 64 kilobytes in length.
- Two link types are supported for the baseband layer : Synchronous Connection-Oriented(SCO)links and Asynchronous Connection-Less (ACL) links. SCQ links support real-time voice traffic using reserved bandwidth. ACL links support best effort traffic. The L2CAP Specification is defined for only ACL links and no support for SCO links is planned.

Frame Format:-

- A frame in the baseband layer can be one of the three types :
 1. One slot
 2. Three slot
 3. Five slot.
- The frame format of the three frame types is shown in Fig. 3.35



This pattern is repeated three times

Fig 3.35: Bluetooth frame format

Access Code

It is a 72 bits field which contains synchronization bits. It also contains the identifier of the master so as to distinguish the frame of one piconet from another.

Header:

- It is a 54 bits field which contains an 18 bits pattern repeated three times. .
- Each such 18 bits pattern consists of the following fields.

Address:

It is a three bit field. So it can define up to seven slaves (1 to 7). The 000 address is reserved for the broadcast communication between a master and the slaves.

Type:

This is a four bit subfield used for defining the type of data, coming from the upper layers.

F: This bit is used for the flow control. F = 1 is an indication of buffer full.

A: This bit is for acknowledgement, since Bluetooth uses the stop-N-wait ARQ.

S: This bit is used to hold the sequence number.

HEC:

This is an eight bit header error correction subfield. It contains the checksum to detect errors in the 18 bit header section.

Data(payload)Field:

This field contains the data or control bits. It can be 0 to 2740 bits long. It contains data or control bits coming from the upper layers.

Security Limitations in Bluetooth:

- Due to the aspect of radio waves, experts fear a security concern with Bluetooth.

- The issue can be addressed with three aspects: specific sequence of channel hopping known only to the sending and receiving devices, challenge-response authentication routine to verify the validity of the receiving unit, and the 128-bit key encryption standard for securing transmission between devices.

Bluetooth Advantages:

- One can create a personal area network at home or on the road with Bluetooth-enabled devices such as keyboard, mouse, scanner, PDA, laptop, cell phone, etc.
- This network can automatically help synchronize notes, calendar, address book and also print pictures, receive emails, access cell phones messages, etc. It can even help consumers pay bills with credit card through Bluetooth cash register if a Bluetooth PDA stores the card information.

3.7 Mobile Generations :-

1G

In this generation of technology, mobile communication is done only through an analog signal that has been used to transmit the user data. The wireless industry has gone through a phase of digitization in which many of the devices that users today use with a wired PC connect to the Internet. 1G technology was primarily used and designed for voice communication purposes. The US Army began using VoIP (Voice over IP) technology in the late 1980s to create a new way of communication for soldiers in wartime, and this first generation of VOIP systems could be considered a breakthrough in the field of communications. However, these systems were not practical as many voice calls were not available, including government communications. It is an analog voice communication system consisting of two microphones, an amplifier, and a signal generator/modulator, called Voice Over Internet Protocol. This method of data transmission enables communication with any other device by using only the telephone line. When the user connects his device via the telephone, he will be connected to the other person over the telephone line. That connection provides both parties with the same information. So, the receiver and transmitter share the information.

2G

The first time the digital signal was used in mobile telephony to transmit signals was in the second generation era of mobile communication. Mobile phones in this generation were capable of transmitting audio and video. The technology was used in more than 60 countries and is already well established in North America and Europe. 2G in a way revolutionized the mobile communication industry although the network speed is limited to 64kbps lots of digital information was being shared.

GPRS technology was introduced it became very useful in transmitting more data through email and other web services that use slightly more network speed. The text message SMS feature in 2G technology was widely used and extremely popular at that time and it allowed many users to occupy a frequency band which made it more efficient.

3G

3G stands for 3rd generation as it just in terms of evolutionary path of the mobile phone industry. In this technology true revolution took place where the network speed was increased up to 2Mbps which made high-speed browsing, gaming, email, and other web services possible to a vast segment of the population. 3G technology has made a scenario where new mobile phones are developed and sold with major advertising capabilities. This makes the users a core demographic and is very important for the mobile service provider. Since all the existing mobile networks now have to run the latest operating systems and hardware, the number of mobile phones has increased from about 600 million in 1998 to nearly 10 billion today. This huge number of mobile phones is changing the face of telecommunications. Today, the mobile operator only has to sell one million smartphones in a year or one percent of the total sales in the mobile industry. This created some downside to the technology where the infrastructure setup for mobile towers and services became high and mobile phones became costly where a most segment of the population couldn't afford to buy. Current 3G speeds are topped out at 14Mbps downlink and 5.8Mbps uplink. 3G technologies use a hybrid of circuit switching and packet switching technology.

4G

4G means 4th generation of mobile phones. 4G is a set of standard that is being developed as future successor of 3G in the very near future. For technology to qualify as 4G technology, it must have speed of up to 100MBPS for moving objects and 1Gbps for stationary user. Another key change in 4G is the abandonment of circuit switching. While 4G uses packet switching only. With 4G, carriers can create a "super high speed" network so that even at 2G speeds it'll be able to deliver fast, reliable downloads in just 30 seconds. The network speed in 4G has been increased up to 100Mbps. This technology made possible some of the things such as high-speed gaming, High-definition Mobile Television, Digital Streaming, HD Television services, Cloud computing, etc. 4G offers faster data speeds at lower prices, in areas that are underserved by its competitors, and offers a wider variety of cellular networks. The development of Smartphones became very popular after the launching of 4G where all the devices manufactured post-2013 supported the 4G feature which made 4G technology reach a wider audience. The 4G technology is being increasingly rolled out in 3G and 3G-LTE networks. As the technology matures, the number of devices and applications to support 4G growth will increase dramatically, from roughly

100 million devices today to over 1.5 billion devices in 2020. 4G offers a lot of capabilities features to the users such as Advanced user capabilities like multitasking, rich content browsing, and instant calling. A new type of multimedia access with Wi-Fi. A new way to upload photos and videos directly from their memory. Ability to share and view online content with friends. Ability to transfer content from the mobile device and use it on the go, etc.

Prof. S. N. Wawale